MARKED-UP SPECIFICATION

Title: APPARATUS AND METHOD FOR PHYSICAL TRANSMISSION OF SPATIALLY-ENCODED OPTICAL DATA

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Abstract

An apparatus and method for the physical transmission of massively parallel, spatially-encoded optical data between an optical emitter array and an optical sensor array. The invention describes two primary embodiments for a photon-native transmission medium that avoids signal conversion bottlenecks. The first embodiment is a flexible "image conduit" cable comprising a coherent bundle of millions of optical fibers, where the spatial arrangement of the fibers is preserved from end to end, directly transmitting an image frame. The second embodiment is a rigid "light-pipe" interconnect using a system of collimating and focusing lenses within a sealed housing to project the image frames over a free-space optical path. Both embodiments provide a physical layer capable of supporting multi-terabit or petabit-per-second data rates by maintaining the inherent parallelism of a frame-based optical data transmission system.

Background of the Invention

The development of massively parallel optical interconnects, which use two-dimensional arrays of emitters and sensors to transmit data, has created a new challenge: the physical medium. These interconnects can generate data at rates of hundreds of terabits per second, far exceeding the capacity of any electrical cable standard like HDMI, DisplayPort, or USB. Converting the massively parallel optical signal into a serial or limited-parallel electrical signal for transmission would re-introduce a severe bottleneck, throttling the performance by orders of magnitude. Therefore, a "photon-native" physical transmission medium is required, capable of conveying millions of parallel data streams as a cohesive, spatially-ordered frame of light over a practical distance.

Summary of the Invention

The present invention provides a physical transmission medium for spatially-encoded optical data, designed to connect a massively parallel optical emitter (e.g., a micro-display) to a massively parallel optical sensor (e.g., a CMOS sensor) without intermediate electronic conversion. The invention is disclosed in two principal embodiments.

The first embodiment, a flexible cable apparatus referred to as an "Image Conduit," is comprised of a coherent fiber optic bundle. A coherent bundle consists of thousands or millions of individual optical fibers that are precisely aligned such that their relative spatial positions are identical at both ends of the cable. When a frame of light is projected onto one face of the bundle, each fiber captures the light from a specific region of the frame and transmits it to the corresponding position on the opposite face, thereby preserving and transmitting the entire image.

The second embodiment, a rigid interconnect apparatus referred to as a "Light-Pipe," is designed for high-fidelity, short-range connections. It consists of a sealed, light-proof housing containing a series of lenses. A collimating lens at the input end captures the light from the emitter array and converts it into a parallel beam. A focusing lens at the output end captures this beam and projects it with precision onto the sensor array. This free-space optical link provides extremely low signal distortion.

Both embodiments provide a physical layer that natively supports the massive parallelism of frame-based optical data transmission, enabling the full multi-terabit potential of the interconnect system.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Government and its agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental and non-commercial purposes.

Brief Description of the Drawings

FIG. 1 is a conceptual diagram illustrating the flexible "Image Conduit" embodiment, showing the coherent fiber optic bundle connecting an emitter and a sensor.

FIG. 2 is a cross-sectional view of the coherent fiber optic bundle, illustrating the preservation of spatial relationships.

FIG. 3 is a conceptual diagram illustrating the rigid "Light-Pipe" embodiment, showing the system of lenses within a sealed housing.

Detailed Description of the Invention

The invention provides two distinct physical apparatus for transmitting frames of spatially-encoded optical data from a transmitter unit to a receiver unit.

Embodiment 1: The Flexible Image Conduit

Referring to FIG. 1, the first embodiment of the system is a flexible cable (100) designed for user-pluggable connections. The system comprises a transmitter module (101) containing an optical emitter array (102) and a receiver module (103) containing an optical sensor array (104). The two modules are connected by the flexible Image Conduit cable (105). This cable is comprised of a coherent fiber optic bundle, meaning the spatial pattern of the fibers is preserved from the input face (106) to the output face (107). When the emitter array (102) projects a data frame, the image is physically transmitted through the bundle to the sensor array (104).

FIG. 2 shows a magnified, schematic cross-section of the coherent fiber bundle (105). A specific fiber at position (X1, Y1) on the input face (106) has its other end at the exact same relative position (X1, Y1) on the output face (107). Likewise, a fiber at a different position (X2, Y2) maintains its relative position across the length of the cable. This one-to-one mapping of spatial locations preserves the integrity of the image frame, ensuring that the data encoded in each "pixel" of light is received at the correct sensor element.

Embodiment 2: The Rigid Light-Pipe

Referring to FIG. 3, the second embodiment is a rigid interconnect (200), suitable for stable, short-range connections such as between servers in a rack or between components on a board. The interconnect comprises a sealed, light-proof housing (201) that connects a transmitter module (101) with an emitter array (102) to a receiver module (103) with a sensor array (104).

Inside the housing, light from the emitter array (102), which is naturally diverging, first passes through a collimating lens (202). This lens converts the diverging rays into a parallel beam of light (203), which travels the length of the housing. At the receiver end, a focusing lens (204)

captures the parallel beam and converges it precisely onto the surface of the sensor array (104). This free-space optical link minimizes signal degradation and is highly robust, though it is sensitive to mechanical shock and misalignment.

~~Claims~~

~~What is claimed is:~~

1. ~~A data transmission apparatus for spatially-encoded optical data, comprising:~~
   - ~~a flexible cable containing a coherent fiber optic bundle, said bundle comprising a plurality of optical fibers; and~~
   - ~~wherein the relative spatial position of each optical fiber in the bundle is maintained from a first end of the cable to a second end of the cable, such that an image projected onto the first end is replicated at the second end.~~
2. ~~The apparatus of claim 1, wherein the cable is configured to optically couple a two-dimensional array of photon emitters at the first end with a two-dimensional array of photon sensors at the second end.~~
3. ~~The apparatus of claim 1, wherein the apparatus is configured to transmit data at a rate exceeding 1 terabit per second.~~
4. ~~A data transmission system, comprising:~~
   - ~~a transmitter module containing a two-dimensional array of photon emitters;~~
   - ~~a receiver module containing a two-dimensional array of photon sensors; and~~
   - ~~a flexible, coherent fiber optic bundle physically connecting the transmitter module and the receiver module, configured to preserve the spatial pattern of photons from the emitter array to the sensor array.~~
5. ~~A method for transmitting data, comprising the steps of:~~
   - ~~generating a two-dimensional frame of light using an array of optical emitters, wherein data is encoded in the spatial characteristics of the frame;~~

- projecting said frame onto a first face of a coherent fiber optic bundle;
- transmitting the frame through the bundle, whereby the spatial arrangement of the frame is preserved; and
- receiving the frame at a second face of the bundle with an array of optical sensors.

6. A rigid data interconnect apparatus, comprising:
   - a sealed, light-proof housing having a first end and a second end;
   - a first port at the first end, configured to receive light from an external two-dimensional array of photon emitters;
   - at least one collimating lens positioned near the first end, configured to convert diverging light from the emitters into a parallel beam;
   - at least one focusing lens positioned near the second end, configured to converge the parallel beam; and
   - a second port at the second end, configured to project the converged beam onto an external two-dimensional array of photon sensors.

7. The apparatus of claim 6, wherein the interconnect establishes a free-space optical path between the first port and the second port.

8. The apparatus of claim 6, wherein the interconnect is configured for short-range communication between electronic components within a server rack.

9. A data transmission system, comprising:
   - a transmitter module containing an array of emitters;
   - a receiver module containing an array of sensors; and
   - a rigid, sealed interconnect optically coupling the transmitter and receiver, said interconnect containing a system of lenses to project frames of light from the emitters onto the sensors.

10. The system of claim 9, wherein the system of lenses comprises at least one collimating lens and at least one focusing lens.

~~[Drawings excised: FIGS. 1, 2, and 3 relocated to standalone formal drawing sheets pursuant to 37 CFR § 1.84]~~